R94 | Pretrained B versus random frozen donor A

Same fresh selector + early node fusion; original R77 four inter-node layers and full loss.

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	0.702263	34129	23890	-0.003073747	-0.030264
VAL	0.429630	1789	1441	-0.000918486	-0.010442
TEST	0.287654	1699	1466	-0.001852840	-0.019016

Split	Target correct	Rescued	Lost	New false	Recall (%)
TRAIN	2730	2359	90	1397	3.2781
VAL	126	115	5	100	2.7243
TEST	117	100	5	101	2.5446

A/B differ only in frozen donor initialization; early fusion + selector freshly initialized identically.

Original CE .35 / comparison .35 / risk .02 / R-Drop .35; neither arm is R93 pure CE.

External pretraining budget unmatched; input coverage16, not16-task pretraining; TSP/ATSP raw.

R50 pool-first is negative evidence. No decoder/routes/new cost labels/MTL restoration.

R89B main / R84A secondary historical. Hardware identities are disclosed on the next page.

ALL deltas equal-weight18 tasks; native FP64 winners, TRAIN SBS; no TEST/LIB model selection.

R94 | Historical VAL screens and hardware

Descriptive locked-model diagnostics, not checkpoint/model reselection.

Arm / reference	Top1 delta pp	Cost delta	Regret delta pp	Pass
R94/A / R89B	-0.249383	0.000903166	0.008845	False
R94/A / R84A	-0.086420	0.000585972	0.006027	False
R94/B / R89B	0.180247	-0.000015320	-0.001597	True
R94/B / R84A	0.343210	-0.000332514	-0.004415	True

A: {"gpu_type": "3090"}

B: {"gpu_type": "4090"}

Hardware differences are disclosed; no claim of bitwise hardware matching.

R94 B vs A | A Inference Timing

First128 locked TEST inputs/task; synchronized GPU wall time, three repetitions; saved evidence only.

Problem	Batch	Donor + selector ms / instance	Cache-hit selector ms / instance
TSP	128	0.403637	0.393278
CVRP	128	0.488799	0.488662
ATSP	128	0.071193	0.070898
OVRP	128	0.073456	0.072977
VRPB	128	0.073150	0.072927
VRPL	128	0.073301	0.072990
VRPTW	128	0.073165	0.073401
OVRPTW	128	0.073171	0.073107
OVRPB	128	0.073385	0.072881
OVRPL	128	0.073080	0.073038
VRPBL	128	0.073631	0.073041
VRPBTW	128	0.072915	0.072817
VRPLTW	128	0.073276	0.073000
OVRPBL	128	0.072864	0.072992
OVRPBTW	128	0.073097	0.072877
OVRPLTW	128	0.073228	0.073196
VRPBLTW	128	0.072942	0.072968
OVRPBLTW	128	0.073125	0.073002

Inputs already on GPU. Excludes dataset I/O and one-time feature-cache preparation.

Cached-feature selector latency is NOT full-system latency; donor+selector is not I/O-inclusive E2E.

TSP/ATSP have unchanged raw paths and no donor; external pretraining budget is unmatched.

Device: cuda:0

R94 B vs A | B Inference Timing

First128 locked TEST inputs/task; synchronized GPU wall time, three repetitions; saved evidence only.

Problem	Batch	Donor + selector ms / instance	Cache-hit selector ms / instance
TSP	128	0.262168	0.261500
CVRP	128	0.317999	0.317428
ATSP	128	0.064552	0.064129
OVRP	128	0.068539	0.068381
VRPB	128	0.068998	0.068472
VRPL	128	0.068735	0.068517
VRPTW	128	0.068802	0.068693
OVRPTW	128	0.068672	0.068616
OVRPB	128	0.068922	0.068748
OVRPL	128	0.068784	0.068531
VRPBL	128	0.069443	0.068753
VRPBTW	128	0.068777	0.068460
VRPLTW	128	0.068952	0.068537
OVRPBL	128	0.068727	0.067991
OVRPBTW	128	0.068671	0.068324
OVRPLTW	128	0.068635	0.068589
VRPBLTW	128	0.068858	0.068624
OVRPBLTW	128	0.068739	0.068639

Inputs already on GPU. Excludes dataset I/O and one-time feature-cache preparation.

Cached-feature selector latency is NOT full-system latency; donor+selector is not I/O-inclusive E2E.

TSP/ATSP have unchanged raw paths and no donor; external pretraining budget is unmatched.

Device: cuda:0